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Digital forensics investigation methodology applicable for social network services

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Abstract Social network services (SNSs) contain various information such as conversations between users, user location information, personal network, and user psychology. This information can be useful for incident investigation. However, in SNSs, unlike computing services that offer services by saving data on a device, a device that uses a SNS with real-time synchronization generally only saves information that is not effective evidence, such as SNS usage log records. However, if digital evidence can be collected through an appropriate digital forensic process, various information such as a social network user's friend list, conversations, and personal relationships can be collected as digital evidence. Therefore, this paper suggests a digital forensic process for digital devices using SNSs. To analyze digital evidence about SNSs, this proposed method is composed of effective processes, classifying digital devices, collecting digital evidence, and analysis.

Keywords Social Network Service · Digital forensic · Data analysis · Mobile device

1 Introduction

The rate of social network service (SNS) use has greatly increased as smart phone distribution has increased. SNSs are becoming the main medium for communication with other service users and presenting oneself publicly. An SNS contains various information such as conversations between users, user location information, personal network, and user psychology. Information like this can be useful evidence that helps incident investigation through digital forensics. However, the SNS provides its service by synchronizing an SNS server and data in real time, unlike a computing service that provides its service by saving data on the device. Hence, when a device on which users access SNS generally collects data, it only saves weak evidence information such as SNS usage log records [2]. However, if digital evidence can be collected through an appropriate digital forensic process, various information such as the SNS

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user's friend list, conversations, and personal connections can be collected as digital evidence. In addition, information saved in the SNS can be collected and be used as evidence.

Therefore, this paper suggests a digital forensic model appropriate for the SNS environment to prevent digital data from damage and manipulation.

2 Related work

2.1 Investigation cases using SNS

– Case with video information

July 2012: the victim was unconscious in the incident and did not remember the sexual assault, but police arrested a group of rapists identified by pictures and videos posted on SNS [7].

– Case with circumstantial evidence

May 2011: in a murder case of a wife of a husband, husband deleted a "Messenger App," conversation record between himself and a woman with whom he had affair. However, police recovered the data and presented the record of the conversation as circumstantial evidence.

– Case with suspect and reason for crime identification

April 2013: in the Boston Marathon terror incident, police collected SNS information (pictures, videos, etc.) of the people at the site to identify suspects. Through this, they inferred the suspects' appearance and clothing. From this SNS information, they also inferred the suspects' political thoughts and the reason for the crime [1, 11].

2.2 Necessity for SNS digital forensics

SNS contains diverse information about users. Such information consists of personal data such as user schedules, conversations with friends, and current location [14]. In a case investigation, the personal data of the suspect saved on an SNS can help in various ways, such as the identification of living patterns, geographical location, and assessment of ideas and mental state. However, it is difficult to collect data useful for a case investigation merely by collecting data from the personal information section (Personal Page) of an SNS user [6].

Some issues include:

– Data privacy settings: the private data of a suspect cannot be accessed from an open account
– Publicly editable data: many users can edit the data of a suspect
– Web browser records that are inefficient for use as digital evidence
– Deletion and modification of records within a mobile device

3 SNS digital forensic service investigation process

To apply digital forensics in an SNS environment, the environment resulting from its provision of a real time synchronization service and openness have to be considered. In addition, because

there are variety of devices that can use SNS, digital forensics procedures should classify devices and consider their abilities [8, 9].

Figure 1 presents an overview of the digital forensic SNS investigation process proposed in this paper.

3.1 Investigation preparation

There is a need for preparation before undergoing digital forensic analysis with a digital device. Such a stage is defined by the digital device and digital forensic.

– Identification of SNSs in use.

When an SNS investigation request is assigned to an investigator, the investigator needs to identify the SNSs used by the suspect. From this, cooperation for the confiscation of the suspect's data can be demanded from the service company providing the SNS.

When the disk on which the data is stored is confiscated during a general digital forensic procedure, the access to the data storage of the SNS should be blocked, similar to using writing protection on a device, so that no modifications can be made [10].

– Acquisition of suspect's account

Among the primary measures of SNS investigation, access information for the suspect's SNS account can be acquired through interrogation. This process allows quick access to the personal information of the suspect and the content created in the SNS.

– Information collection using SNS

After the identification of the SNS used by the suspect and the acquisition of the suspect's account information, the investigator starts to collect information about the type of SNS the

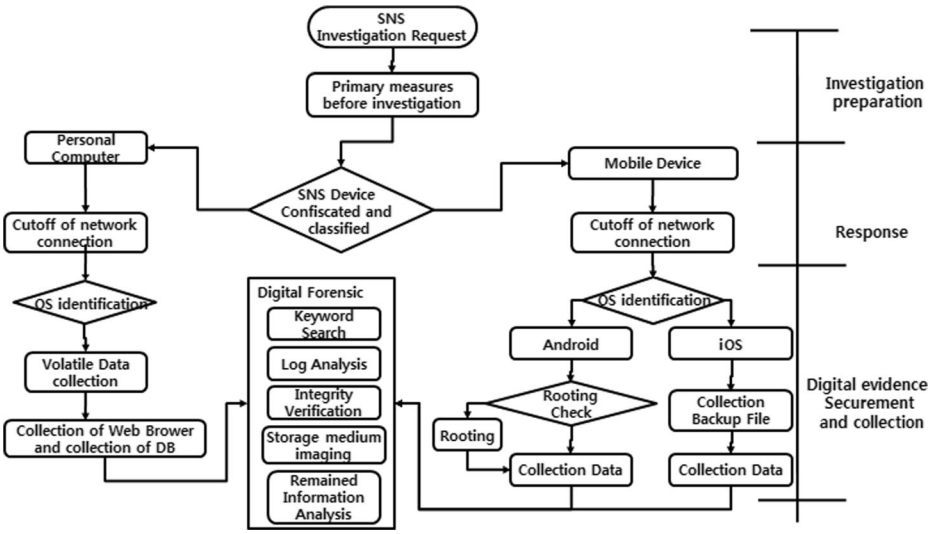

Fig. 1 SNS digital forensic investigation procedure

suspect uses and his account. This process collects information about the device before digital forensics such as SNS-use timeline, networking, and suspect location are compared and analyzed using the gathered digital information [3].

3.2 Response

In a case investigation, the first action taken by an investigator in the field is called the Response. This is conducted when the investigator arrives at the site after preparing for the investigation with the procedure outlined above.

– Digital device categorization

The detective collects the devices with which the suspect accesses SNS and categorizes them as PCs or mobile devices [15]. This is because the treatment process later on is different for these two devices. To ensure integrity, a single detective cannot collect a digital device. The collection of evidence is recorded with a camera and camcorder, and the originals and copies are both secured using forensic duplication equipment. This treatment ensures the integrity of the digital evidence.

– Network connection cutoff

The devices collected by the investigator must be disconnected from the network to prevent any modification of information stored about SNS. This is because an SNS is a network service provided in real time from data stored in the SNS server via a network, not a storage-type service that provides a service from information stored inside the local digital device.

– Suspect's account suspension

By suspending the suspect's SNS account, data can be conserved. During a typical digital forensic process, if the disc on which the data are saved is confiscated, this blocks access to the file space in SNS similarly to the use of writing prevention equipment and ensures that the suspect or other users are unable to change the data in the suspect's SNS file space.

The network connection blocking process is similar to blocking the networks of PC and mobile device but there is a difference in the method.

3.3 Digital evidence securement and collection

Imaging for the purpose of evidence analysis must be done for the conservation of original data and further analysis when digital evidence is collected. Imaging the storage medium, a process that prevents damage to the original data, is an important initial stage of the digital forensic procedure.

The equipment or software used in the storage medium image process prints a hash value as it creates a copy of the original evidence or image. This must be recorded in the process of evidence collection to secure integrity and reliability.

3.4 Delivering evidence and confirmation

The transportation of digital evidence is a significant issue because digital evidence can be easily damaged or modified. Until the moment of submission of the original as evidence to

an analysis office, it is critical to make the effort to be careful so that the copy is transported safely.

3.5 Investigation and analysis

The procedure for the securement and collection of digital evidence is similar to standard investigations. It adopts the existing overall digital forensic process, while each procedure is adapted according to the characteristic of SNS.

4 Digital evidence investigation techniques

4.1 Personal computer (PC)

4.1.1 Real time data investigation procedure

The information that forms the evidence in this paper is the volatile data of a live system. To use volatile data that is as likely to be damaged. To perform such an analysis, the device must be confiscated and the memory analysis must be conducted after imaging the live system information with a memory dump that creates a single image file out of system memory.

– Experimental Environment

  OS: Windows 7 Ultimate K
  CPU: Intel® Core i7 2.93 GHz
  RAM: 4.0 GB
  S/W: MDD (Memory DD), HxD (Hex Editor)

Figure 2 shows a screenshot of the memory dump of a live system's physical memory to the file "memory dump.mdd."

```
C:₩Users₩treasure₩Desktop>mdd_1.3.exe -o c:₩Memorydump.mdd
 -> mdd
 -> ManTech Physical Memory Dump Utility
    Copyright (C) 2008 ManTech Security & Mission Assurance

 -> This program comes with ABSOLUTELY NO WARRANTY; for details use option `-w'
    This is free software, and you are welcome to redistribute it
    under certain conditions; use option `-c' for details.

 -> Dumping 3319.36 MB of physical memory to file 'c:₩Memorydump.mdd'.


 849756 map operations succeeded (1.00)
 0 map operations failed

 took 66 seconds to write
 MD5 is: f24629489cb66e80cbe1cdad0fd58876
```

Fig. 2 Memory dump process

Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start
0x87a39030	iexplore.exe	5560	2360	13	573	1	0	2013-09-10 05:35:03
0x85b1daf8	iexplore.exe	1164	5560	29	1003	1	0	2013-09-10 05:35:04
0x85cc7030	chrome.exe	6140	2360	40	1179	1	0	2013-09-10 05:35:28
0x85d68030	chrome.exe	6060	6140	13	315	1	0	2013-09-10 05:35:32
0x85d74458	chrome.exe	2764	6140	13	238	1	0	2013-09-10 05:35:37
0x8868dd40	jucheck.exe	4624	2640	6	405	1	0	2013-09-10 05:35:38
0x85b42540	chrome.exe	2444	6140	16	282	1	0	2013-09-10 05:35:45
0x85bc4708	chrome.exe	1544	6140	24	295	1	0	2013-09-10 05:36:04

Fig. 3 Memory use list showing web browser use

4.1.2 Real time data collection procedure

1. Process list extraction

To analyze the imaged memory of the live system with respect to SNS data, one must look for traces of SNS use in memory. Computer devices use a web browser to use SNS, unlike mobile devices that use an application [4, 5]. That is why, to analyze the trace of SNS use, one must find the memory address of the web browser. In this study, Volatility, a memory analysis tool, is used to extract the process list from the dumped memory image. Figure 3 shows the memory offset value and operation times of process IDs (PIDs) and parent process IDs (PPIDs). This live system shows the use of Chrome and Internet Explorer.

Figure 3 shows the operation time, memory offset value, process ID (PID) and the parent process ID (PPID) of web browsers in use in the dumped live system memory use list. This live system shows the use of Chrome and Explorer web browser

2. Memory imaging for process

In order to analyze the memory in use by a web browser, the memory of each individual web browser process must also be imaged.

Figure 4 shows image dumps of the assigned process memories of PIDs 5560, 1164, 6140, and 6060 of Chrome and Internet Explorer.

```
C:₩volatility-2.1>python vol.py -f c:₩test₩Memory.mdd -p 1164 memdump --dump-dir
 c:₩test₩1₩ --profile=Win7SP1x86
Volatile Systems Volatility Framework 2.1
************************************************************************
Writing iexplore.exe [  1164] to 1164.dmp

C:₩volatility-2.1>python vol.py -f c:₩test₩Memory.mdd -p 6140 memdump --dump-dir
 c:₩test₩1₩ --profile=Win7SP1x86
Volatile Systems Volatility Framework 2.1
************************************************************************
Writing chrome.exe [  6140] to 6140.dmp

C:₩volatility-2.1>python vol.py -f c:₩test₩Memory.mdd -p 6060 memdump --dump-dir
 c:₩test₩1₩ --profile=Win7SP1x86
Volatile Systems Volatility Framework 2.1
************************************************************************
Writing chrome.exe [  6060] to 6060.dmp
```

Fig. 4 Imaging web browser processes

```
2764.dmp
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
08798ED0  34 36 33 36 2D 38 35 46 45 2D 41 41 45 34 35 37  4636-85FE-AAE457
08798EE0  42 44 46 45 45 38 68 74 74 70 73 3A 2F 2F 77 77  BDFEE8https://ww
08798EF0  77 2E 66 61 63 65 62 6F 6F 6B 2E 63 6F 6D 2F 70  w.facebook.com/p
08798F00  68 6F 74 6F 2E 70 68 70 3F 76 3D 35 30 32 30 34  hoto.php?v=50204
08798F10  38 37 38 36 34 38 32 38 38 39 26 73 65 74 3D 76  8786482889&set=v
08798F20  62 2E 34 38 31 31 35 33 33 39 31 39 30 35 37 36  b.48115339190576
08798F30  32 26 74 79 70 65 3D 32 26 74 68 65 61 74 65 72  2&type=2&theater
08798F40  68 74 74 70 73 3A 2F 2F 77 77 77 2E 66 61 63 65  https://www.face
08798F50  62 6F 6F 6B 2E 63 6F 6D 2F 70 68 6F 74 6F 2E 70  book.com/photo.p
08798F60  68 70 3F 76 3D 35 30 32 30 34 38 37 38 36 34 38  hp?v=50204878648
```

Fig. 5 Trace of SNS use in a web browser

4.1.3 Real time data analysis

In the dump files that have been imaged from each web browser process, there is volatile data resulting from web browser operation. This volatile data is data that cannot be modified by the suspect or anyone else in an SNS. Moreover, even after the deletion of data shared in an SNS by the suspect or others, there is a copy of the data remaining in this memory. In such a case, it can provide useful information in the investigation through real time data analysis.

Real time data analysis can be done by searching for SNS-related data in the memory data stored in the web browser process image dump files.

Figure 5 shows a trace of SNS use by process PID:2764 in a memory image dump file of a Chrome web browser.

– SNS user networking information

Through real time SNS data analysis of volatile memory, discovering the network of a suspect is possible. When a computer device is used with a web browser, one can search for the profiles of SNS users that the suspect visited.

Figure 6 shows the profile address of an SNS user that was visited. It was found by searching for the web addresses of Facebook user profile pages in the process memory image dump file of a Chrome web browser.

```
6140.dmp
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
0042C8D0  63 63 65 70 74 3A 20 2A 2F 2A 0D 0A 52 65 66 65  ccept: */*..Refe
0042C8E0  72 65 72 3A 20 68 74 74 70 73 3A 2F 2F 77 77 77  rer: https://www
0042C8F0  2E 66 61 63 65 62 6F 6F 6B 2E 63 6F 6D 2F 70 72  .facebook.com/pr
0042C900  6F 66 69 6C 65 2E 70 68 70 3F 69 64 3D 31 30 30  ofile.php?id=100
0042C910  30 30 33 37 39 32 36 39 35 37 38 35 26 66 72 65  003792695785&fre
0042C920  66 3D 70 62 26 68 63 5F 6C 6F 63 61 74 69 6F 6E  f=pb&hc_location
0042C930  3D 66 72 69 65 6E 64 73 5F 74 61 62 0D 0A 41 63  =friends_tab..Ac
0042C940  63 65 70 74 2D 45 6E 63 6F 64 69 6E 67 3A 20 67  cept-Encoding: g
0042C950  7A 69 70 2C 64 65 66 6C 61 74 65 2C 73 64 63 68  zip,deflate,sdch
0042C960  0D 0A 41 63 63 65 70 74 2D 4C 61 6E 67 75 61 67  ..Accept-Languag
0042C970  65 3A 20 6B 6F 2D 4B 52 2C 6B 6F 3B 71 3D 30 2E  e: ko-KR,ko;q=0.
```

Fig. 6 Visited users' profile IDs

6140.dmp | 2444.dmp

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

0109C0B0  03 00 00 00 F0 03 00 00 66 6F 72 20 28 3B 3B 29  ....ð...for (;;)
0109C0C0  3B 20 7B 22 74 22 3A 22 6D 73 67 22 2C 22 73 65  ; {"t":"msg","se
0109C0D0  71 22 3A 32 36 31 2C 22 6D 73 22 3A 5B 7B 22 65  q":261,"ms":[{"e
0109C0E0  76 65 6E 74 22 3A 22 72 65 61 64 22 2C 22 74 69  vent":"read","ti
0109C0F0  64 73 22 3A 5B 22 69 64 2E 34 36 32 32 37 30 30  ds":["id.4622700
0109C100  30 37 31 38 37 36 32 39 22 5D 2C 22 63 68 61 74  07187629"],"chat
0109C110  5F 69 64 73 22 3A 5B 31 30 30 30 30 33 33 35 33  _ids":[100003353
0109C120  37 34 38 35 30 39 5D 2C 22 6D 61 72 6B 5F 61 73  748509],"mark_as
0109C130  5F 72 65 61 64 22 3A 74 72 75 65 2C 22 66 6F 6C  _read":true,"fol
0109C140  64 65 72 5F 69 6E 66 6F 22 3A 7B 22 69 64 2E 34  der_info":{"id.4
0109C150  36 32 32 37 30 30 30 37 31 38 37 36 32 39 22 3A  62270007187629":
0109C160  22 69 6E 62 6F 78 22 7D 2C 22 74 69 6D 65 73 74  "inbox"},"timest
0109C170  61 6D 70 22 3A 31 33 37 38 37 39 32 30 35 32 31  amp":13787920521
0109C180  32 35 2C 22 74 79 70 65 22 3A 22 6D 65 73 73 61  25,"type":"messa
0109C190  67 69 6E 67 22 7D 2C 7B 22 65 76 65 6E 74 22 3A  ging"},{"event":
0109C1A0  22 72 65 61 64 22 2C 22 74 69 64 73 22 3A 5B 22  "read","tids":["
0109C1B0  69 64 2E 34 36 32 32 37 30 30 30 37 31 38 37 36  id.4622700071876
0109C1C0  32 39 22 5D 2C 22 63 68 61 74 5F 69 64 73 22 3A  29"],"chat_ids":
0109C1D0  5B 31 30 30 30 30 33 33 35 33 37 34 38 35 30 39  [100003353748509
```

Fig. 7 Record of conversation of users

com.facebook.katana	2013-05-27	00:12
app_acra-reports	2013-05-21	20:34
app_dex	2013-05-21	20:34
app_minidumps	2013-05-21	20:34
app_outdex	2013-05-21	20:34
app_secondary-program-dex-jars	2013-05-21	20:34
app_secondary_program_dex	2013-05-21	15:55
app_secondary_program_dex_opt	2013-05-21	20:34
cache	2013-05-22	00:32
databases	2013-05-27	09:52
analytics_db2	36864 2013-05-27	09:12
bookmarks.db	26624 2013-05-22	03:46
contacts_db2	1160192 2013-05-27	07:58
fb.db	51200 2013-05-27	00:12
graphql	6144 2013-05-22	07:31
newsfeed_db	125952 2013-05-22	12:45
notifications.db	256000 2013-05-27	09:05
pages_db	5120 2013-05-22	05:35
prefs_db	38912 2013-05-27	09:52
qe_db	9216 2013-05-27	09:12
threads_db2	83968 2013-05-27	06:12
timeline_db	5120 2013-05-22	09:32
webview.db	14336 2013-05-22	00:43
webviewCache.db	192512 2013-05-22	00:14
dex-lock-dummy	1 2013-05-27	07:31
files	2013-05-22	00:32
lib	2013-05-21	12:29
shared_prefs	2013-05-27	09:52

Fig 8 SNS database file

– Information related to SNS user conversations.

One can analyze conversations between the suspect and other SNS users from the data stored in volatile memory. In case of conversations using a Web browser, the acquisition of any evidence is difficult, as users can easily delete it. However, the memory stores the content of a conversation. Hence, as long as the computer device is not shut down, a conversation record still remains that can be evidence.

Figure 7 shows the content of a conversation between profile ID:10003356748509 and the user of the targeted device at time 13792052125. It was collected by searching for the "msg" string. Since the web browser records the time as the number of microseconds elapsed since 00 h 00 min 00 s of January 1, 1601, it is expressed as 13792052125

– Timeline information of SNS user

One can analyze not only the networking information of an SNS user and data related to an SNS conversation, but also the timeline of SNS accesses using a web browser. Comparing such information with cache information and web browser history enables the investigation of records deleted by the user or others.

4.2 Mobile device

To effectively apply digital forensics to the mobile devices that access SNS, the real time synchronization environment that is provided by the SNS and openness should be considered.

– Experimental Environment

OS: Android 4.12
CPU: Quad-Core Snapdragon 600 1.7 GHz
RAM: 2.0 GB
S/W: SQLite Database Browser

4.2.1 SNS service mobile device data analysis methods

SNS use on a mobile device is enabled through an SNS application installed on the mobile device [12]. Applications like these pull a variety of information from the SNS server when logging in to provide good services after saving that information to its database. A digital forensic detective can analyze information useful for investigation through the information in those databases.

This study analyzed the application database of the current Facebook. Figure 8 shows the database files located in the data/data/com.facebook.katana/database of a mobile device. Database files like these enable fast service for a user by receiving data from SNS while connected in real time and saving them.

The application database files of interest to a digital forensic detective conserve the data at the time of confiscation because the network connection is blocked after the mobile device is secured.

Files and content that can be used to extract information from database files are as shown in Table 1.

Table 1 Database file information

File name	Summary	Information
fb	Saving information of a user	uid : user ID, secret : coded hash value, username : user mail address, machine_id : user machine ID, access_token : access token value, session_key : session connection hash value, profile : user name https://fbcdn-profile-a.akamaihd.net//hprofile-ak-ash4//273835_100002545740576_1019605470_q.jpg : user picture
contacts_database 2	User's friend information	contactId : friend connection relationship ID, profileFbid : friend's ID value, graphApiWriteId : friend relationship, name : friend's name, smallPictureUrl : https://fbcdn-profile-a.akamaihd.net/hprofile-ak-snc6/c32.9.117.117/s100x100/190686_103259136424781 _7050306_a.jpg : friend's profile picture. communicationRank : closeness, displayNumber : mobile phone number
threads_database2	User's conversation information	Saving user's conversation contents
analytics_database2	Mobile device Form used	GPS check, network used, country accessed from, mobile battery status, etc.

5 Expected Advantages

The SNS real time data analysis method discussed in this paper enables an effective investigation in the SNS environment.

Table 2 is a table that shows the possibility of finding analyzable digital evidence according to analysis environment. Through this table it can be seen that the digital forensics of memory analysis are not greatly affected by network environment or data deletion compared to the information that can be gathered from existing Web or mobile devices.

The following advantages can be expected when using this forensic SNS procedure.

- Preparation for any confusion in an investigation process by the systemization of an investigation procedure for SNS.
- Assurance of the expedited collection of evidence and its integrity with appropriate response in accordance of environment of SNS for the collection of digital evidences.

Table 2 Search availability of SNS digital evidence according to environment

Classi fication	Network connection			Network cutoff			Deletion of general data		
	Web	Mobile	Memory	Web	Mobile	Memory	Web	Mobile	Memory
Human networking	O	O	O	X	X	O	X	O	O
Location data	O	O	O	X	X	O	X	X	O
Closed post	X	X	O	X	X	O	X	X	O
Secret information	X	X	O	X	X	O	X	X	O

X: Searching is impossible O: Searching is possible

– Systematic collection of digital evidence through the relevant analysis of various devices, files, and web browser information [13].

6 Conclusion

An SNS contains a variety of information such as the user's profile information, location information, political thoughts, and personal network. However, unlike a computing service where services are provided through data saved in the device, the service is provided by synchronizing with data in real time. This is why computer devices only save weak evidence information. Thus, this paper proposed a digital forensic process appropriate for SNSs that uses digital data in an SNS for incident investigation evidence. For this, an SNS digital forensic procedure was newly constructed and effective evidence investigation techniques to collect digital SNS data was analyzed. Digital data analysis was done after categorizing devices into either PCs or mobile devices. The SNS digital forensic process suggested in this paper included some additional processes in the currently used common digital forensic process and a mobile forensic process. Because of this, it can be applied without an overall change in the procedure. By applying the suggest method in this essay, it is expected that SNS information can be effectively used for incident investigations.

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